INCH-POUND

MIL-C-48684A (AR)
17 February
SUPERSEDING
MIL-C-48684 (AR)
1 April 1986

MILITARY SPECIFICATION

CARTRIDGE, 40MM, M918 TP LOADING, ASSEMBLING AND PACKAGING

This Specification is approved for use by the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packaging for one type of cartridge designated as Cartridge, 40mm, M918 TP.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited 'n 'he solicitation (see 6.2) .

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-60942 - Primer Percussion for 40mm Ammunition

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1310

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP-Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawing and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those *in* effect on the date of solicitation.

DRAWINGS (see 6.11)

Us. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKING DRAWINGS

8886405 - Linking of 40mm High Velocity Ammunition
9399372 - Cartridge, 40mm, M918TP
9362543 - Packing and Marking of Shipping and Storage Container M548 with Linked 40mm Cartridges
12928042 - Packing and Marking of Shipping and Storage Container PA120 with linked 40mm Cartridges

INSPECTION EQUIPMENT DRAWINGS

9202253 - Chamber Gage
9202254 - Limit
9202255 - Alignment
9202929 - Action Time

(Copies of other Government documents, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing In this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing (Dwg.) 9399372, all associated drawings and with all requirements specified in applicable specifications and drawings.

3.3 X-ray examination of projectile assembly prior to assembly to cartridge case assembly. The projectile assembly shall be x-rayed for improper assembly, missing or damaged parts. missing operations, foreign material and an armed or partially armed fuze.

3.4 X-ray examination of ballistic sample. Prior to forwarding the cartridges to the proving ground for ballistic testing, they shall be subjected to x-ray examination.

3.5 Functioning. The cartridge assembly shall function without casualty and comply with the following:

3.5.1 Velocity. The average velocity of the sample cartridges, conditioned at 70°F and fired in the MK19, MUD III, Grenade Machine Gun shall be 790 ± 10 feet per second (fps) with a standard deviation of 12.0 fps or less.

3.5.2 Action time and base plug movement. When fired in the M129 Grenade Launcher, the action time of the cartridge, conditioned at 70°F, shall not exceed 4 milliseconds and there shall be no base plug movement more than .005 above flush with rear of cartridge case after firing.

3.5.3 Impact functioning. When fired from a MK19 MOD III, Grenade Machine Gun the projectile assembly upon impact shall function satisfactorily as evidenced by smoke, flash and sound. There shall be no evidence of an air burst (see 6.9).

3.5.4 Safety. When fired from the MK19, MOD III, Grenade Machine Gun the projectile shall not lodge in the bore of the weapon and there shall be no evidence of a premature burst (see 6.10) .

3.6 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough and workmanlike mariner. They shall be free of burrs, chips, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however a few light scratches not exposing base material may be permitted. All marking and stamping shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract *or* purchase *order,* the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government *reserves* the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General inspections. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality Conformance Inspection (4.4)

4.3 First article inspection. Not applicable

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each inspection lot shall contain:

a. Cartridge case assembly metal parts from one lot interfix number and from one manufacturer.

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b. Primers from one lot interfix number and from one manufacturer.

c. Projectile assemblies from no more than two (2) consecutively produced serial lots and from one manufacturer.

d. Propellant from not more than one (1) lot number and from one manufacturer.

e. Only cartridges from one lot per ammo belt/can shall be used.

f. Links from one lot interfix number and one manufacturer.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table I below, using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE I. Attributes sampling inspection

Lot Size	Inspection Level		
	III	IV	V
2 to 8	*	*	5
9 to 15	*	13	5
16 to 25	*	13	5
26 to 50	32	13	5
51 to 90	32	13	13
91 to 150	32	13	13
151 to 280	32	32	20
281 to 500	32	32	20
501 to 1200	80	50	20
1201 to 3200	80	50	32
3201 to 10000	125	50	32
10001 to 35000	125	80	50
35001 to 150000	125	80	50
150001 to 500000	200	125	50
500001 and over	200	125	50

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Note: Numbers under inspection levels indicate sample size; Asterisks indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero failures and reject on one or more failures for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.13) . When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-48684A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge Case Assembly, Prior to Loading	1 OF 1	8886327
			NEXT HIGHER ASSEMBLY: 9399372

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Closing cup missing	100%	3.2	Visual/Probe
MAJOR				
101.	Propellant weight		3.2	4.5.2
102.	Propellant weight (alternate method)		3.2	4.5.3
MINOR	None defined			

NOTES:

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-48684A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cartridge Case Assembly, Prior to insertion of Base Plug and Primer	1 OF 1	8886327
			NEXT HIGHER ASSEMBLY: 9399372

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Spilled propellant from cartridge case	100%	3.2	Visual
2.	Propellant charge weight obviously incorrect (See 6.8)	100%	3.2	Visual
MAJOR				
101.	Propellant charge weight obviously incorrect (See 6.8)	100%	3.2	Visual
MINOR	None defined			

NOTES:

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-48684A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Cartridge Case Assembly	1 OF 1	8886327
			NEXT HIGHER ASSEMBLY: 9399372

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Primer above flush with base plug	100%	3.2	Gage
Major				
101.	Security of crimp of base plug and action time	1/ 2/	3.2 3.5.2	4.5.4 9202929
102.	Base plug above flush	Level III	3.2	Gage
103.	Depth of base plug, max	Level III	3.2	Gage
104.	Depth of primer from base plug, max.	Level III	3.2	Gage
105.	Primer damaged	Level III	3.2	Visual
106.	Excessive sealing compound on exterior surface of primer	Level III	3.2	Visual
Minor				
201.	Poor Workmanship	Level V	3.6	Visual

NOTES:

1/ Five (5) cartridge case assemblies shall be randomly selected every four (4) hours from each manual primer assembly machine or ten (10) cartridge case assemblies shall be randomly selected every two (2) hours from each automatic primer assembly machine.

2/ If any base plug moves more than the applicable requirement or if any action time is greater than the applicable requirement, the two (2) hour or four (4) hour production quantity represented shall be rejected.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-48684A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Projectile Assembly, prior to assembly to Cartridge Case Assembly	1 OF 1	9399374
			NEXT HIGHER ASSEMBLY 9399372

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	X-ray examination 1/	100%	3.3	4.5.5
MAJOR				
101.	X-ray examination 2/	100%	3.3	4.5.5
MINOR				
201.	X-ray examination 2/	100%	3.3	4.5.5

NOTES:

1/ A critical defect is defined as a setback pin not adjacent to the rotor gear, setback pin not seated in bottom of counterbore in escapement housing, setback pin or spring missing, detent spring or detent missing, detent not engaged in teeth of rotor gear, or pinion or verge assembly missing.

2/ A major or minor defect is defined as any defect not specified in Note 1/.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-48684A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Projectile Assembly, prior to assembly to Cartridge Case Assembly			9399374
				NEXT HIGHER ASSEMBLY
				9399372
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	O-ring (packing) missing from groove	Level III	3.2	Visual
MINOR	None defined			

NOTES:

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-48684A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2	DRAWING NUMBER
4.4.2.6	Cartridge, 40mm, M918TP			9399372
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1. 1/	X-ray examination of ballistic sample	288(M548), 256(PA120),0-1	3.4	4.5.6
MAJOR				
101.	Chamber gage failure	100%	3.2	9202253 9202254 9202255
102.	Total Length, maximum	100%	3.2	Gage
103. 2/	X-ray examination of ballistic sample	288(M548) 3/ 256(PA120)	3.4	4.5.6
104.	Air pressure test	4.5.7	3.2	4.5.7
105.	Pull test of projectile	4/	3.2	4.5.8
106.	Gap, maximum, at rear of rotating band	Level III	3.2	Gage
107.	Packing exposed	Level III	3.2	Visual

NOTES:

1/ A critical defect is defined as any defect listed in Note 1/ of 4.4.2.4.
2/ A major or minor defect is defined as any defect not specified in Note 1/.
3/ Major or minor defects found during inspection shall be counted against the sample that was selected for detection of that particular defect. Any sample containing a known defect shall not be fired. 4/ Select one (1) sample from each (30) minutes of production. If the sample fails the drawing requirement, the thirty (30) minutes of production represented by the sample shall be rejected.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-48684A (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.6	Cartridge, 40mm, M918TP	SHEET 2 OF 2	9399372
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
MINOR				
201. 2/	X-ray examination of ballistic sample	288(M548) 3/ 256(PA120)	3.4	4.5.6
202.	Marking missing, improper or illegible	Level V	3.2	Visual
203.	Poor Workmanship	Level V	3.6	Visual
	Functioning Tests			
	Functioning			
5/	MK19, MOD III, machine gun - single shot	144(M548) 5/ 128(PA120)	3.5, 3.5.1, 3.5.3, 3.6,	4.5.9.1
5/	MK19, MOD III, machine gun, rapid fire	144(M548)6/5/ 128(PA120)6/5/ 48(M548)7/5/ 32(PA120)7/5/	3.5,	3.5.3,

NOTES:

5/ See Table II for defect classification and acceptance/rejection criteria.

6/ Three (3) full metal cans (48 linked cartridge per can for M548) or four (4) full metal cans (32 linked cartridges per can for PA120) shall be randomly selected.

7/ When ten (10) consecutive lots pass all rapid fire requirements, reduced testing of one (1) full metal can (48 linked cartridges per can for M548 and 32 linked cartridges per can for PA 120) shall be instituted. If any lot fails to meet any rapid fire requirement, testing of three (3) M548 full metal cans (or four (4) PA 120 full metal cans) shall be reinstated until ten (10) lots again pass this requirement.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-48684A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Linked Ammunition, Prior to Packing	1 OF 1	8886405

NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Linked improperly (anyone of the four (4) tabs of link not in groove; link inverted; link damaged; link distorted)	100%	3.2	Visual
MAJOR	None defined			
MINOR	None defined			

NOTES:

AMSMC [illegible] 1570a-E, 1 Jul 89 Replaces A[illegible]C Form 1570a, 1 Apr 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-48684A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Unsealed Shipping and Storage Container	1 OF 1	12928042 or 9362543
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Cartridge inverted	Level III	3.2	Visual
102.	Incorrect number of cartridges	Level III	3.2	Visual
103.	Fillers missing	Level III	3.2	Visual
104.	Support, Top or Bottom missing (when Applicable)	Level III	3.2	Visual
105.	Spacer, missing	Level III	3.2	Visual
MINOR	None defined			

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-48684A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Shipping and Storage Container	1 OF 1	12928042 or 9362543
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
Major				
101.	Box or container damaged	Level III	3.2	Visual
MINOR				
201.	Contents loose	Level V	3.2	Manual
202.	Can seal missing or improperly positioned	Level V	3.2	Visual
203.	Marking misleading or unidentifiable	Level V	3.2	Visual

NOTES:

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TABLE II - QUALITY CONFORMANCE INSPECTION

FUNCTIONING & DEFECT CLASSIFICATION

AND ACCEPTANCE/REJECTION CRITERIA

MK19, MOD III, GRENADE MACHINE GUN

	Acceptance Number		Rejection Number		Classification Defect
	MK19 Single Shot	MK19 Rapid Fire	MK19 Single Shot	MK19 Rapid Fire	
1. Failure to meet Average velocity (see 3.5.1)	1/	--	2/	--	Major
2. Standard deviation of velocity over max. (see 3.5.1)	1/	--	2/	--	Major
3. Impact functioning failure (see 3.5.3)	10 (M548) 3/ 8 (PA120)		11 (M548) 3/ 9 (PA120)		Major
4. Primer leak as evidenced by perforation of primer cup or smoke deposit on base plug or cartridge case (see 3.5)	0	0	1	1	Major
5. Projectile lodges in weapon bore (see 3.5.4)	0	0	1	1	Critical
6. Premature burst (see 3.5.4)	0	0	1	1	Critical
7. Airburst (see 3.5.3)	0	0	1	1	Major
8. Primer misfire (see 3.5)	1	1	2	2	Major

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TABLE II - QUALITY CONFORMANCE INSPECTION

FUNCTIONING & DEFECT CLASSIFICATION

AND ACCEPTANCE/REJECTION CRITERIA

MK19, MOD III, GRENADE MACHINE GUN

	Acceptance Number		Rejection Number		Classification Defect
	MK19 Single Shot	MK19 Rapid Fire	MK19 Single Shot	MK19 Rapid Fire	
9. Weapon stoppage excluding primer misfire (see 3.5)	0	0	1	1	Major

NOTES:

1/ Lot acceptable if requirement is met.

2/ Lot rejected if requirement is not met.

3/ For informational purposes only.

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4.4.4 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment design shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Check test for deterioration for primers. If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-60942 immediately before the primer lot is assembled into the cartridge. This test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into the cartridge (see 6.6).

4.5.2 Propellant weight in cartridge case. The propellant weight shall be determined and then check weighed 100 percent. The check weighing shall be accomplished independently of the original weighing for determination, using a different balance from that used to make the original weighing and if performed manually, shall be performed by another operator. Any charge which falls to comply with the requirement specified on the applicable drawing shall be classified defective and removed from the lot.

4.5.3 Propellant weight of case, cartridge alternate method. During production, every propellant charge shall be weighed or volumetrically loaded. Those failing to comply with the requirements specified by the applicable drawing shall be classified defective and removed from the lot. At the start of production, 380 consecutive samples per station, per machine, shall be selected and checked weighed 100 percent. All samples shall comply with the applicable drawing requirement. If all samples comply with the applicable drawing requirement, 5 samples per station, for every 2 hours production, shall be selected and the weight of each of the 5 samples must meet the applicable drawing requirement. If any weight fails the applicable drawing requirement, the 2 hours production represented by that sample shall be rejected. Correction will be made and 380 consecutive samples check weighed from that station. If all samples meet the applicable requirement, sampling every 2 hours shall be resumed.

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4.5.4 Security of crimp of base plug and action time. Time cartridge assembly shall be fired in a M129 Grenade launcher. Observation shall be made for base Plug movement and action time. The movement of the base plug shall be measured from the highest point of the base plug to a point on the cartridge case adjacent to it.

4.5.5 X-ray examination of projectile assemblies. Prior to assembly to the cartridge case assembly, the projectile assembly shall be x-rayed for critical, major and minor defects. The test shall be conducted in two (2) planes. One shall be from the top of the projectile down and the other from the side.

4.5.6 X-ray examination of ballistic sample. Prior to the forwarding of the cartridges to the proving ground for ballistic testing, the cartridges shall be X-rayed *for* critical, major and minor defects. The test shall be conducted in two (2) planes. One shall be from the top of the cartridge down and the other from the side.

4.5.7 Air pressure. The cartridge assembly shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. Fourteen (14) samples shall be selected from every thirty (30) minutes of production. The last sample shall be air tested. If it is acceptable, the thirty (30) minutes of production will be considered Acceptable; and the samples shall be returned to the lot. If the sample is unacceptable, the remaining thirteen (13) samples shall be air tested. If no failure occurs, the thirty (30) minutes of production shall be considered acceptable. if one (1) or more failures occur, the thirty minutes of production shall be rejected and segregated from the lot.

4.5.8 Pull test of projectile. The cartridge shall be placed in an approved fixture and the axial tension specified on the applicable drawing shall be applied. Cartridge case assemblies may be reused. The projectile shall be pulled until separation occurs and data recorded.

WARNING: This test is conducted on assemblies containing *an* explosive element. Test barricades (where used), procedures and equipment shall have prior approval.

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4.5.9 Functioning. This test shall be conducted at a Government owned proving ground. The cartridge being tested shall be temperature conditioned at 70° ± 5°F for a minimum of eight (8) hours, and then fired within fifteen (15) minutes after being removed from temperature soak. The weapon used shall have been “broken in"n if new with at least forty (40) cartridges. At the start of the functioning test, six (6) gun warmers shall be fired.

4.5.9.1 MK19, MOD III, grenade machine gun, single shot. The linked cartridges shall be fired single shot and impact at 200 meters ±5 meters from the weapon into 10 inches, minimum, of coarse sand. The cartridges shall be fired at a rate not to exceed four per minute. The sand shall be raked so that individual craters do not overlap. In addition, the sand shall be raked after each dud. If a projectile assembly falls outside the impact medium, it shall be declared a ‘no-test”. The velocity shall be measured at 30 feet from the muzzle of the machine gun. The monitoring devices and the recorders shall be a dual system setup for recording two (2) separate readings and shall be within 3 fps of each other. If they are not within 3 fps the results shall be discarded, the cartridge declared "no-test” and another fired in its place. If a weapon is suspected of being the cause of rejection for the mean velocity and standard deviation phase, sufficient cartridges from a controlled lot shall be fired from the same approved weapon. If there is a significant difference in the standard deviation and mean velocity of the control lot, then the lot shall be rejected. Observation shall be made for functioning at the impact medium and any premature burst or air burst. Velocity shall be recorded. The standard deviation of velocity shall be calculated. The expended cartridge case shall be examined for primer leakage. When the proving ground reports ballistic test results, the number of "no-test" cartridges shall also be recorded.

4.5.9.2 MK19, MOD III grenade machine gun, rapid fire. The linked cartridges shall be fired in bursts of five (5), and” impact at 200 meters ± 5 meters from the weapon into 10 inches, minimum, of coarse sand. The sand shall be raked so that individual craters do not overlap. In addition, the sand shall be raked after each dud. If a projectile assembly falls outside the impact medium, it shall be declared a “no-test”. Velocities of the cartridge shall not be recorded. Observation shall be made for functioning at the impact medium and any premature bursts or airburst. The expended cartridge cases shall be examined for primer leakage. When the proving ground reports ballistic test results, the number of “no-test” cartridges shall also be recorded.

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5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. Preservation and packaging shall be in accordance with Dug. 12928042 or 9362543 as applicable.

5.2 Packing.

5.2.1 Level A. The linked cartridge assembly shall be packed in accordance with Dwg. 12928042 or 9362543 as applicable.

5.3 Marking. Marking shall be in accordance with Dug 12928042 or 9362543 as applicable.

5.4 Shipping. When shipments of more than one lot are made at one time, such as in a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

5.5 Palletizating. Palletization shall be in accordance with AMC Drawing 19-48-4232/21-20 PM 1007 for the PA120 container and Drawing 19-48-4116/26A-20PA 1002 for the M548 container.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the metal parts and loading, assembling and packing for Cartridge, 40MM, TP, M918.

6.2 Acquistion requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Applicable stock number.

d. Packaging requirements, if other than specified in Section 5.

e. Serialization requirement, if applicable.

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6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, U.S. Army ARDEC, Attn: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of results of contractor conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examination or tests to Commander, U.S. Army ARDEC, Attn: SMCAR-QAF-S, Picatinny Arsenal, NJ 07806-5000.

6.5 Submission of test data. In addition to the' normal distribution of records, when the cartridges are procured by AMCCOM, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, U.S. Army ARDEC, Attn: SMCAR-QAF-S, Picatinny Arsenal, NJ 07806-5000.

6.6 Cost of check test. The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the primer assemblies into the cartridge and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer assembly lot was under the jurisdiction of either contractor (when applicable).

6.7 Standard deviation shall be calculated from the following formula:

$$S = \sqrt{\frac{\sum (X - \bar{X})^2}{n-1}}$$

where: X = each individual value

$\bar{X}$ = sample arithmetic mean $\frac{\sum X}{n}$

n = sample size

$\sum (X - \bar{X})^2$ = the sum of the squares of the differences between the sample mean and each individual value.

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6.7.1 For a faster and easier method of arriving at the standard deviation formula, the following alternate method may be used:

$$s = \sqrt{\frac{n\sum X^2 - (\sum X)^2}{n(n-1)}}$$

6.8 Propellant weight. If any cartridge case, prior to insertion of the base plug and *primer. is* suspected of containing an underweight *or pverweght* propellant charge, based on visual inspection, it shall be removed from the lot and weighed on a suitable scale. Any cartridge case found with less than 75 percent *or* more than 125 percent *of* the assessed propellant weight will be classified as a critical defect and removed from the lot. Any cartridge case found to have a proellant weight out of drawing tolerance but within ± 25 percent of the assessed propellant weight shall be classified as a major defect and removed from the lot.

6.9 Air burst. A projectile that functions in flight further than 20 feet from the weapon muzzle.

6.10 Premature burst. A premature burst is considered to have occurred if the projectile functions in the weapon bore or in flight up to 20 feet from the weapon muzzle.

6.1 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.12 Subject term (Key word) listing.

Target Practice (TP)
MK19 MODIII Grenade Machine Gun
M548 Shipping and Storage Container
PA120 Shipping and Storage Container

6.13 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

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6.14 Changes from previous issue. Marginal Notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-AS43)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-A-48684A (AR)	940217

3. DOCUMENT TITLE: CARTRIDGE, 40MM, M918 TP LOADING, ASSEMBLING AND PACKAGING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. REMARKS

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS *(Include Zip Code)* ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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